



EUROPEAN UNIVERSITY OF LEFKE - "Faculty of Engineering"

"Computer Engineering"

SYLLABUS

2024-2025 Spring Semester

Course Code	Course Title	Course Type	Weekly Course Hours			Credits	ECTS	Weekly Time Schedule
			T	A	L			
EE431	Digital Image Processing	Elective	3	0	0	3	5	Thursday 15:00 - 17:50
Prerequisite		Prerequisite to						
Course Lecturer	Asst. Prof. Dr. Cem Kalyoncu					Office Hours Schedule		
E-mail	ckalyoncu@eul.edu.tr							
Phone	2527					Office / Room No	AS-308	
Teaching Assistant						Phone		
E-mail						Office / Room No		
Catalogue Descriptions	This course introduces the principles of digital image processing applications and their implementations mainly in C++. Topic covers: Image sampling and quantization; interpolation techniques, nearest neighbor interpolation, bilinear interpolation; Histograms, understanding image histogram, constrast stretching, brightness and contrast, gamma, histogram equalization. Filtering in spatial domain, low pass filter, high pass filter, band pass filter, box filter, edge detection techniques. Color theory, human color vision, digital image color systems: RGB, HSI, HSV, CMYK. Image morphology, thresholding, erosion, dilation, opening and closing operations, and/or/not operations. Information theory, Shannon'e entropy, Huffmann compression, compression techniques, lossy/lossless compression.							
Course Objectives	At the end of this course the student should be able to implement and perform experiments using generic image processing algorithms as well as being able to understand and implement published methods. At the end of the course, students will be well equipped to perform research on image processing topics.							
Learning Outcomes	1. Familiarity with image processing terms 2. Knowledge of histograms and histogram processing 3. Ability to understand and perform filters on grayscale and binary images 4. Ability to understand and implement image processing algorithms 5. Having theoretical understanding of topics related to image processing, such as color theory, information theory							
Programme Outcome Relations	PO1: 3 PO2: 5 PO3: 4 PO4: 5 PO5: 2 PO6a: 1		PO7: 1 PO8: 2 PO9: 2 PO10a: 1 PO10b: 1 PO11: 2			(1) Strongly disagree; (2) Disagree; (3) Neither agree nor disagree; (4) Agree; (5) Strongly agree.		
Textbooks and/or References	1	R. C. Gonzalez and R. E. Woods, Digital Image Processing, 3rd Edition, Prentice Hall, 2008						
	2							
	3							
WEEK	Date	TOPICS						Reference No - Section
Week 1		Introduction, relation with signal processing, course outline and rules						-
Week 2		Images, continuous and discrete images, sampling, quantization, image acquisition.						Ch 1, 2
Week 3		Simple image transformations, interpolation and interpolation methods						Ch 2
Week 4		Hands on training, image processing in C++						-
Week 5		Histograms, histogram processing, contrast stretching, histogram equalization						Ch 3
Week 6		Practical examples, Quiz 1						-
Week 7		Filtering in spatial domain, convolution, smoothing, sharpening, edge detection.						Ch 3, 5
Week 8		Frequency domain, frequency domain filters, low pass, high pass, and band pass filters						Ch 4, 5
Week 9		Color theory, color models, color based segmentation						Ch 6
Week 10		Midterm						
Week 11		Morphology, dilation, erosion, opening, closing						Ch 9
Week 12		Morphology examples, Quiz 2						
Week 13		Information theory, image compression, huffman coding						Ch 8
Week 14		Advanced topics in image processing: super resolution, contours, object detection, image enhancement techniques, video coding						-
Week-15/16		Finals						
Evaluation Tools	Evaluation Tool		Quantity		Date		Weight in Total (%)	Weight in Semester Evaluation (%)
	Final Exam		1				40	
	Semester Evaluation						60	
	Midterm(s)		1				30	50.0
	Quiz(zes)		2				10	16.7
	Project(s)							
	Homework(s)		4				20	33.3
	Laboratory works							
Attendance								
*** Lifelong Learning Programme (LLP) ***					Language of Instruction:		English	
Evaluation Tool		Quantity	Student Workload Hours	Evaluation Tool		Quantity	Student Workload Hours	
Theoretical lecturing hours (TLH)		13	39	Homework		4	20	
TLH self study		13	39	Project				
Quiz (Q)		2	2	Presentation				
Q preparation self study		2	10	Seminar				
Midterm exam (ME)		1	2	Final exam (FE)		1	2	
ME preparation self study		1	10	FE preparation self study		1	20	
TOTAL :								144
Recommended ECTS Credit (Total Hours / 30) :								4.80